5th LATAM School - The Latin American School in Software Engineering. Co-located with CBSOFT, XVI Brazilian Conference on Software: Practice and Theory, September 22nd to 26th | Recife/PE, Brazil

EVALUATING THE QUALITY OF ARTIFICIAL-INTELLIGENCE-GENERATED CODE IN FIXING TECHNICAL DEBT AND MERGE CONFLICTS

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Introduction

Merge conflicts and **technical debt** (TD) challenge modern software development [1].

AI tools like GitHub Copilot are widely used but **not rigorously evaluated** for these tasks.

The goal of this research project is to **understand the role AI-generated code plays in collaborative software development**.

To achieve this goal, we need to understand:

- What are the **most common merge conflicts** found by software developers?
- What are the **most frequent technical debts** needed to be addressed by software developers?
- Is AI-generated code useful for
  - Resolving **merge conflicts**?
  - Addressing **simple project issues**?
  - Addressing **design project issues**?
  - Addressing **project vulnerability issues**?
  - Increasing **test-coverage**?
  - Addressing Internet of Things **(IoT) issues**?

Background and Related Work

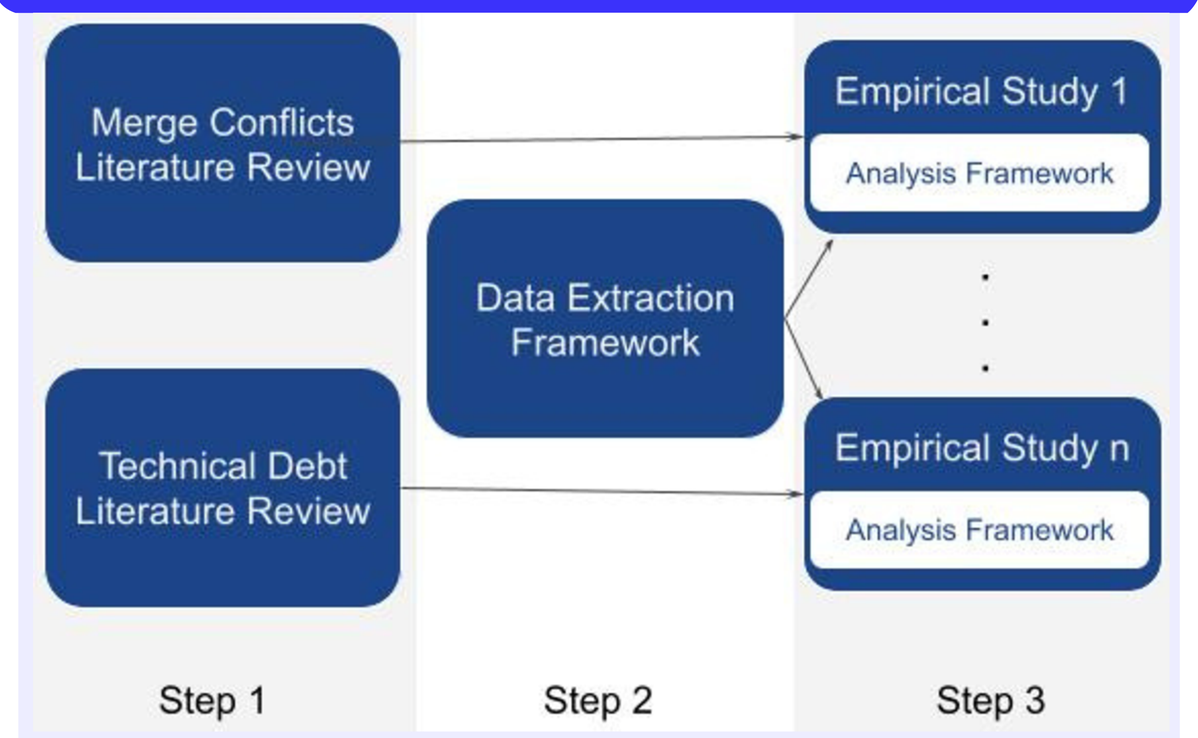
Researchers' concerns have changed over the years following **tools to support software development** [2].

Although the TD metaphor was introduced over three decades ago, it has only garnered significant attention from researchers in the past ten years [3].

Empirical studies have revealed that while **AI tools can be effective in producing correct and low-complexity code**, they also occasionally **introduce security vulnerabilities or lower code quality** [4, 5].

AI tools hold significant promise in reshaping the development process, hence, research is required to fully understand their impact on code quality, maintainability, and security.

Materials and Methods



Expected Outcomes and Conclusion

By the end of the project, we expect to deliver **practical guidelines and recommendations for developers on how to best leverage AI tools** to enhance their workflows, minimize technical debt, and improve software reliability and security.

Through our interdisciplinary collaboration, we will ensure that the **findings** are robust, applicable, and **beneficial to both academia and industry**.

The open dissemination of our results, tools, and datasets will further enhance the impact of this research, making the findings accessible to a broad audience and fostering future innovations.

References

[1] M. Joblin, S. Apel, and W. Mauerer. "**Evolutionary trends of developer coordination: a network approach.**" In: Empirical Software Engineering (EMSE) 22.4, pp. 2050–2094, 2017.
[2] Gustavo Vale. **Investigating the Merge Conflict Life-Cycle Taking the Social Dimension into Account**. Doctoral Dissertation, Saarland University, p. 264, 2024.
[3] N. Brown, Y. Cai, Y. Guo, R. Kazman, M. Kim, P. Kruchten, E. Lim, A. MacCormack, R. Nord, I. Ozkaya, R. Sangwan, C. Seaman, K. Sullivan, and N. Zazworka. **Managing technical debt in software-reliant systems**. In Proceedings of the FSE/SDP workshop on Future of software engineering research (FoSER). ACM, pp.47–52, 2010.
[4] D. OBrien, S. Biswas, S. M. Imtiaz, R. Abdalkareem, E. Shihab, and H. Rajan, "**Are prompt engineering and todo comments friends or foes? an evaluation on github copilot**" in Proceedings of the IEEE/ACM 46th International Conference on Software Engineering (ICSE), pp. 1–13, 2024.
[5] N. Nguyen and S. Nadi, "**An empirical evaluation of github copilot's code suggestions**" in Proceedings of the 19th International Conference on Mining Software Repositories (MSR), p. 1–5, 2022.